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Advanced Control System Design - Tutorial 5 – Solution

1. Consider the following system:

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\\ \dot{x}_3\end{bmatrix}=\begin{bmatrix}0 & 1 & 0\\ 0 & 0 & 1\\ 20 & 16 & 1\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix}+\begin{bmatrix}0\\ 8\\ 24\end{bmatrix}u$$

$$y=\begin{bmatrix}1 & 0 & 0\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix}$$

A. Investigate the state controllability of the system using different approaches.

Solution:

Using the standard formula:

$$|\mathbf{M}|=\begin{vmatrix}\mathbf{B} & \mathbf{AB} & \mathbf{A}^2\mathbf{B}\end{vmatrix}=\begin{vmatrix}0 & 8 & 24\\ 8 & 24 & 152\\ 24 & 152 & 696\end{vmatrix}=0\Rightarrow$$ Not completely state controllable.

Using inspection (must be transformed to the diagonal or Jordan form first):

$$|\lambda\mathbf{I}-\mathbf{A}|=\begin{vmatrix}\lambda & -1 & 0\\ 0 & \lambda & -1\\ -20 & -16 & \lambda-1\end{vmatrix}=\lambda^3-\lambda^2-16\lambda-20=(\lambda+2)(\lambda+2)(\lambda-5)$$

$$\lambda_1=-2,\ \lambda_2=-2,\ \lambda_3=5$$

$$\mathbf{S}=\begin{bmatrix}1 & 0 & 1\\ \lambda_1 & 1 & \lambda_3\\ \lambda_1^2 & 2\lambda_1 & \lambda_3^2\end{bmatrix}=\begin{bmatrix}1 & 0 & 1\\ -2 & 1 & 5\\ 4 & -4 & 25\end{bmatrix}$$

$$\dot{\mathbf{z}}=\mathbf{S}^{-1}\mathbf{A}\mathbf{S}\mathbf{z}+\mathbf{S}^{-1}\mathbf{B}u$$

$$\begin{bmatrix}\dot{z}_1\\ \dot{z}_2\\ \dot{z}_3\end{bmatrix}=\begin{bmatrix}0.918 & -0.082 & -0.020\\ 1.429 & 0.429 & -0.143\\ 0.082 & 0.082 & 0.020\end{bmatrix}\begin{bmatrix}0 & 1 & 0\\ 0 & 0 & 1\\ 20 & 16 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & 1\\ -2 & 1 & 5\\ 4 & -4 & 25\end{bmatrix}\begin{bmatrix}z_1\\ z_2\\ z_3\end{bmatrix}$$
$$+\begin{bmatrix}0.918 & -0.082 & -0.020\\ 1.429 & 0.429 & -0.143\\ 0.082 & 0.082 & 0.020\end{bmatrix}\begin{bmatrix}0\\ 8\\ 24\end{bmatrix}u$$

$$\begin{bmatrix}\dot{z}_1\\ \dot{z}_2\\ \dot{z}_3\end{bmatrix}=\begin{bmatrix}-2 & 1 & 0\\ 0 & -2 & 0\\ 0 & 0 & 5\end{bmatrix}\begin{bmatrix}z_1\\ z_2\\ z_3\end{bmatrix}+\begin{bmatrix}-1.143\\ 0\\ 1.143\end{bmatrix}u \Rightarrow$$ Not completely state controllable.

Using the transfer function approach:

$$G(s)=\mathbf{C}(s\mathbf{I}-\mathbf{A})^{-1}\mathbf{B}+D$$

$$G(s)=\begin{bmatrix}1 & 0 & 0\end{bmatrix}\begin{bmatrix}s & -1 & 0\\ 0 & s & -1\\ -20 & -16 & s-1\end{bmatrix}^{-1}\begin{bmatrix}0\\ 8\\ 24\end{bmatrix}+0$$

$$G(s)=\begin{bmatrix}1 & 0 & 0\end{bmatrix}\left[\frac{1}{s^3-s^2-16s-20}\begin{bmatrix}s^2-s-16 & s-1 & 1\\ 20 & s(s-1) & s\\ 20s & 4(4s+5) & s^2\end{bmatrix}\right]\begin{bmatrix}0\\ 8\\ 24\end{bmatrix}$$

$$G(s)=\frac{8\cancel{(s+2)}}{\cancel{(s+2)}(s+2)(s-5)}$$

Not completely state controllable due to the zero-pole cancellation

B. Investigate the output controllability of the system.

Solution:

$$\mathbf{N}=\begin{bmatrix}\mathbf{CB} & \mathbf{CAB} & \mathbf{CA^2B} & \mathbf{D}\end{bmatrix}=\begin{bmatrix}0 & 8 & 24 & 0\end{bmatrix}$$

The rank is 1. Thus, the system is completely output-controllable.

C. Determine whether the system is stabilizable or not.

Solution:

From the solution of A:

$\lambda_1 = -2,\ \lambda_2 = -2,\ \lambda_3 = 5$

Thus, $\mathbf{T}_1 = \mathbf{T}_2$.

$$\mathbf{T}_1 = \mathbf{T}_2 = \left[\lambda_1\mathbf{I} - \mathbf{A} \quad \mathbf{B}\right] = \left[\begin{bmatrix} -2 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{bmatrix} - \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 20 & 16 & 1 \end{bmatrix} \quad \begin{matrix} 0 \\ 8 \\ 24 \end{matrix}\right] = \begin{bmatrix} -2 & -1 & 0 & 0 \\ 0 & -2 & -1 & 8 \\ -20 & -16 & -3 & 24 \end{bmatrix}$$

The rank is 2. Thus, the modes ($\mathbf{T}_1$ and $\mathbf{T}_2$) are uncontrollable, but stable.

$$\mathbf{T}_3 = \left[\lambda_1\mathbf{I} - \mathbf{A} \quad \mathbf{B}\right] = \left[\begin{bmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{bmatrix} - \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 20 & 16 & 1 \end{bmatrix} \quad \begin{matrix} 0 \\ 8 \\ 24 \end{matrix}\right] = \begin{bmatrix} 5 & -1 & 0 & 0 \\ 0 & 5 & -1 & 8 \\ -20 & -16 & 4 & 24 \end{bmatrix}$$

The rank is 3. Thus, the unstable mode ($\mathbf{T}_3$) is controllable.

Thus, the system is stabilizable.